Project Initialization and Planning Phase

Date	31 October 2026
Team ID	Individual Project (No Team ID)
Project Title	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report	
Objective	Develop practical knowledge of Tableau and data analytics by analysing global fuel prices, taxation, subsidies, and economic patterns through comparative visual analysis and interactive dashboards.
Scope	The project uses the supplied global_fuel_prices_Dataset.csv dataset and Tableau Public 2026.3. It covers country and regional comparisons, taxation and subsidy patterns, income-level comparisons, and historical fuel-price trends. It does not provide live fuel prices, forecasts, or proof of causal relationships.
Difficulty comparing global fuel prices with taxation, subsidy levels, regions, income groups, and changes over time using a raw dataset alone.	
Description	The dataset contains thousands of records across multiple countries, regions, dates, and fuel-related measures. Without organized visualizations, users may find it difficult to compare groups, identify patterns, and interpret the information consistently.
Impact	A clear visual analysis can make cross-country and regional comparisons easier and support exploratory learning and discussion related to energy policy, supply-chain planning, and economic research. Observed associations should not be treated as proof of causation.
A published Tableau Public workbook containing three dashboards: (1) Fuel Taxation & Subsidy Policy, (2) Global Fuel Prices & Supply Chain Planning, and (3) Fuel Prices, Taxation & Economic Development, supported by a historical trend worksheet.	
Approach	Connect the CSV dataset to Tableau Public, use the available date, country, region, income_level, subsidy_level, petrol, diesel, LPG, Brent crude, and tax fields, build comparative charts and filters, arrange them into three dashboards, and publish the workbook for viewing.
Key Features	Interactive country and region comparisons; top and bottom petrol-price rankings; petrol/diesel comparisons; taxation-versus-price analysis; subsidy-level comparisons; income-level comparisons; and fuel-price trends alongside Brent crude. The workbook uses the supplied dataset and its available fields.

Resource Requirements

Resource Type	Description	Specification/Allocation
Exact device model/specifications were not recorded in the project documentation.		
Computing Resources	Processing is handled through Tableau Public; no dedicated GPU requirement was specified.	No specific CPU/GPU allocation recorded; use a computer capable of running a modern web browser.
Memory	Memory available on the user device.	Exact RAM not recorded; no project-specific RAM allocation documented.
Storage	CSV dataset and Tableau workbook files.	Dataset: global_fuel_prices_Dataset.csv; approximately 27,468 records. Local disk capacity not recorded.
Tableau Public 2026.3; modern web browser.		
Frameworks	No separate application framework used.	Not applicable; dashboards built directly in Tableau.
Libraries	No separate programming/data-analysis libraries used.	Not applicable; Tableau built-in aggregations, charting, and calculated fields where used.
Development Environment	Dashboard design, analysis, and publication environment.	Tableau Public 2026.3; published workbook accessible through Tableau Public.
global_fuel_prices_Dataset.csv; CSV format; approximately 27,468 records, 84 countries, 7 regions; January 2020–April 2026.		
Data	Source, size, format	e.g., Kaggle dataset, 10,000 images

Project Design Phase-I
Problem – Solution Fit Template

Date	31 October 2026
Team ID	Individual Project (No Team ID)
Project Name	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Maximum Marks	2 Marks

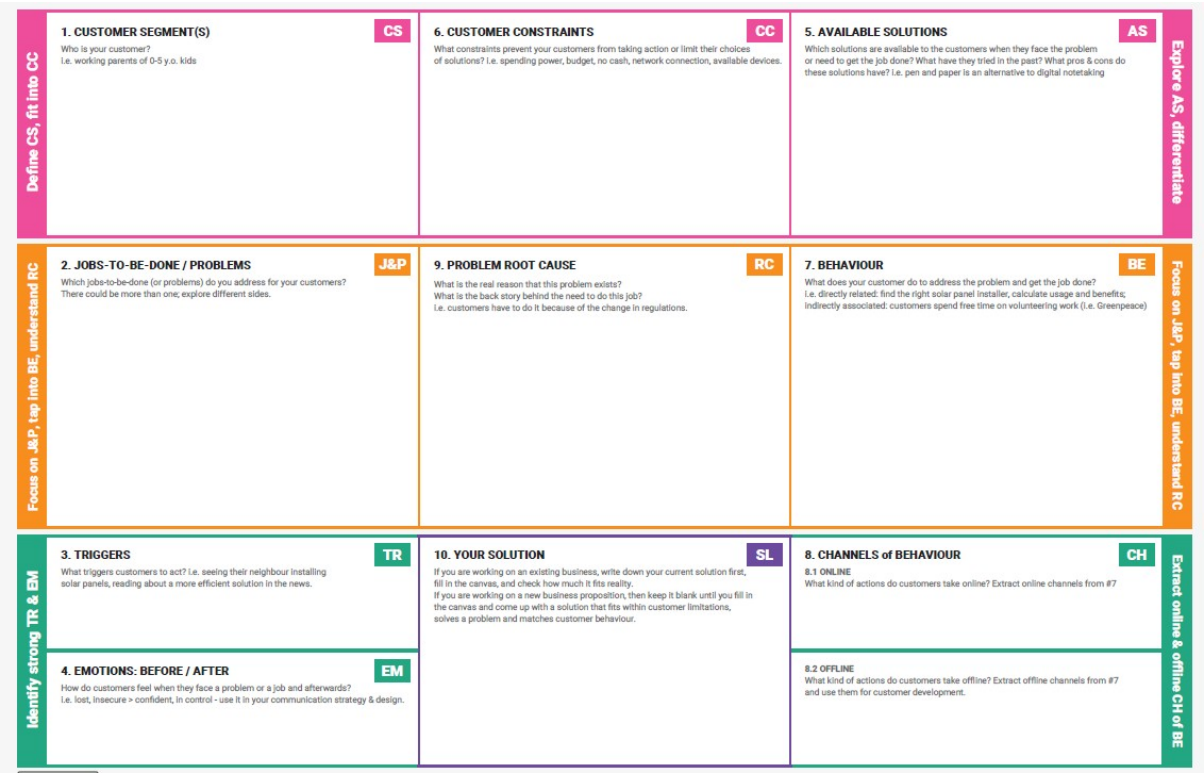
Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Template:



References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe

Completed Problem–Solution Fit (Project-Specific)

This completed analysis applies the original canvas to the three intended dashboard user groups. These are project-defined analytical personas and inferred needs, not findings from customer interviews or formal user testing. The dashboard is an exploratory visualization tool; observed relationships do not establish causation.

1. Customer Segment(s)

Energy policy consultants; global supply-chain managers; energy economics researchers, students, and other users comparing fuel-price and taxation patterns.

2. Jobs-to-be-Done / Problems

Compare fuel taxes and petrol prices across countries and regions; identify high- and low-price countries for supply-chain context; explore differences in fuel prices and taxation by subsidy and income level; review historical fuel-price movements alongside Brent crude.

3. Triggers

A need to prepare a policy comparison, assess fuel-cost differences across markets, explain regional variation, or explore how prices and taxation vary over time.

4. Emotions: Before / After

Before: users may feel overwhelmed by many countries, dates, and indicators in a raw dataset. After: users may find comparisons easier to explore and explain through organized charts, rankings, and filters. These are anticipated reactions, not measured results.

5. Available Solutions

Raw CSV/spreadsheet review, individual country statistics, published reports, and other data visualizations. These may require manual comparison across multiple views or sources.

6. Customer Constraints

Users may have limited time, varying data-literacy levels, limited familiarity with Tableau, screen-size constraints, and questions about data coverage or interpretation. The dataset has defined fields and a date range, so conclusions are limited to the available records.

7. Behaviour

Open the published Tableau workbook, choose a relevant dashboard, read chart titles and legends, compare countries or categories, apply available filters, and interpret patterns with care.

8. Channels of Behaviour

Online: Tableau Public workbook viewed in a web browser and shared workbook link. Offline: users may discuss or present observations in class, meetings, or reports. These are intended usage channels, not observed user behaviour.

9. Problem Root Cause

Fuel-price, tax, subsidy, country, region, income-level, and date information can be difficult to compare consistently when viewed as a large table. Different scales and category groupings can also make patterns less immediately apparent.

10. Your Solution

A Tableau Public workbook built from global_fuel_prices_Dataset.csv, with three dashboards: (1) Fuel Taxation & Subsidy Policy, (2) Global Fuel Prices & Supply Chain Planning, and (3) Fuel Prices, Taxation & Economic Development. Visuals include country rankings, regional comparisons, taxation-versus-petrol-price views, subsidy and income-level comparisons, and a Brent crude/fuel-price trend view. The workbook supports exploration; it does not prove causal relationships or provide real-time prices.

Fit Summary and Limitations

The solution fits the identified analytical tasks by consolidating relevant measures into interactive comparisons and trend views. Fit is currently reasoned from the project requirements and available dashboard features. It has not been validated through interviews, surveys, or formal usability testing. The workbook is based on the provided dataset and should be interpreted within its coverage and field definitions.

Project Design Phase-I
Solution Architecture

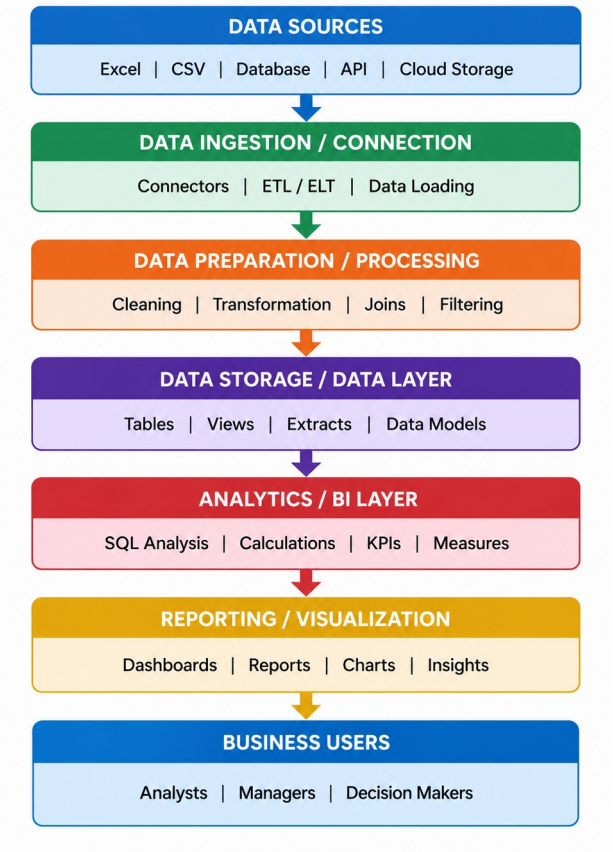
Date	31 October 2026
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Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used

Project-Specific Solution Architecture

The following architecture describes the solution actually used for this project. The dataset is a CSV file connected directly to Tableau Public; no separate database, Snowflake environment, external ETL pipeline, or live price feed is claimed.



Figure: Architecture and data flow for the Global Fuel Prices and Taxation Trends project.

Component Description Table — Project-Specific

Component Name	Description / Role in Architecture	Technology Used
Data source	Historical country-level observations including fuel prices, tax percentage, region, subsidy level, income level and date.	global_fuel_prices_Dataset.csv
Data connection	Connects the CSV dataset to the Tableau workbook.	Tableau Public data connection
Data preparation	Uses Tableau data-source settings, field roles/types, filters and built-in aggregations as needed for the views. No external cleaning tool is claimed.	Tableau Public 2026.3
Analysis and business logic	Uses chart-level measures, groupings, filters and calculated fields only where configured in the workbook.	Tableau worksheets / calculations
Visualization layer	Contains the worksheets for tax, petrol/diesel, subsidy, income-level and Brent crude comparisons.	Tableau worksheets
Dashboard layer	Organizes the analysis into three scenario-based dashboards for policy, supply-chain and economic-development perspectives.	Tableau dashboards
Publishing and access	Publishes the workbook for browser-based exploration; users can interact with available filters and legends.	Tableau Public
Users	Students, analysts, policy-oriented users and supply-chain/economic researchers can explore historical patterns.	Web browser

Scope and limitations

- The workbook analyzes the historical dataset supplied for the project (approximately January 2020 to April 2026, as described in the project dataset notes). It does not automatically fetch current pump prices or update itself from an external API.
- The charts support comparison and exploratory analysis. Associations between taxation, subsidies, income levels, crude oil and fuel prices should not be presented as proof of causation.
- The three dashboards are published in Tableau Public. The additional worksheet “Global Petrol Price Trend” is retained in the workbook but is not one of the three dashboards.

Dashboard Design

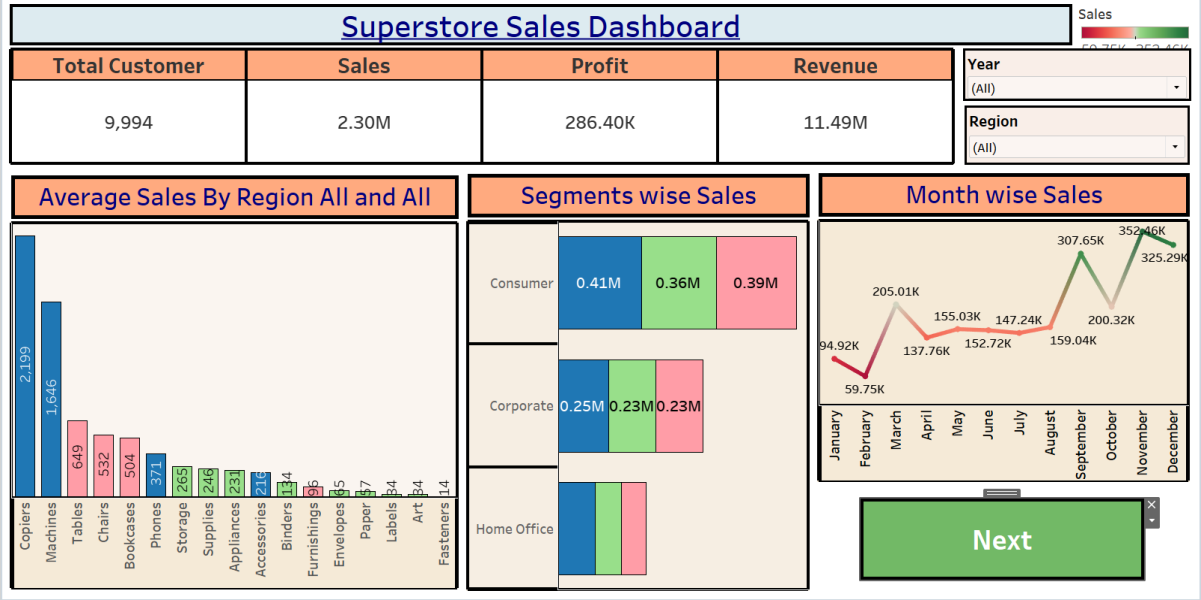
Date	31 October 2026
Team ID	Individual Project (No Team ID)
Project Name	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Maximum Marks	5 Marks

Creating an effective dashboard involves thoughtful design to ensure that the presented information is clear, relevant, and easily understandable for the intended audience. Here are some key principles and best practices for dashboard design

Activity 1: Interactive and visually appealing dashboards

Creating interactive and visually appealing dashboards involves a combination of thoughtful design, effective use of visual elements, and the incorporation of interactive features. Here are some tips to help you design dashboards that are both visually appealing and engaging for users so take care of below points

- Clear and Intuitive Layout
- Use Appropriate Visualizations
- Colour and Theming
- Interactive Filters and Slicers
- Drill-Down Capabilities
- Responsive Design
- Custom Visuals and Icons
- Use of Infographics



Note: Highlight the major outcomes in form of bullet points

Sample:

Here are five potential outcomes from the dashboard image provided:

- **Customer Count and Sales Overview**:The dashboard reports a total of 9,994 customers, with cumulative sales of 2.30M, a profit of 286.40K, and revenue of 11.49M. These figures highlight the overall scale and profitability of the store's operations.
- **Highest Sales**: Copiers and Machines are the top-selling products, with sales values of 2.199M and 1.646M, respectively. This suggests strong demand or high price points for these products.
- **Peak Month**: December has the highest monthly sales at 352.46K, likely due to holiday shopping.
- **Seasonal Sales Growth**:The monthly sales chart shows growth in specific months, such as March, September, October, and December, likely influenced by seasonal factors and sales events.
- **Sales Distribution Across Segments**: The **Consumer segment** leads with 0.41M in sales, followed by Corporate at 0.25M, and Home Office at the lowest

Activity 2

Publish Dashboard on Tableau Public and Paste the Dashboard Public link below:

Example

https://public.tableau.com/views/SalesDashboard_17247660582000/Dashboard1?:language=en-

US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

COMPLETED DASHBOARD DESIGN — TABLEAU PROJECT

This section documents the three dashboards implemented in Tableau Public. The screenshots below are the latest dashboard images supplied by the project owner. Descriptions reflect the visible elements in those screenshots.

Dashboard 1 — Scenario 1: Fuel Taxation & Subsidy Policy

Analyse fuel prices, taxation, and subsidies to support exploratory energy-policy comparisons.

Layout and visualisation components

Chart / component	Visual type	Design purpose
Top 10 Countries by Average Fuel Tax Percentage	Horizontal bar chart	Compare countries by average fuel-tax percentage.
Average Petrol Price by Region (USD/Liter)	Horizontal bar chart	Compare average petrol prices across seven regions.
Impact of Taxation on Petrol Prices	Scatter plot	Explore the relationship between average tax percentage and average petrol price; this does not establish causation.
Average Petrol Prices by Subsidy Level	Bar chart	Compare average petrol prices across subsidy categories.

Interactivity and layout

A Region legend supports interpretation of regional colours. The dashboard uses a two-by-two chart grid.

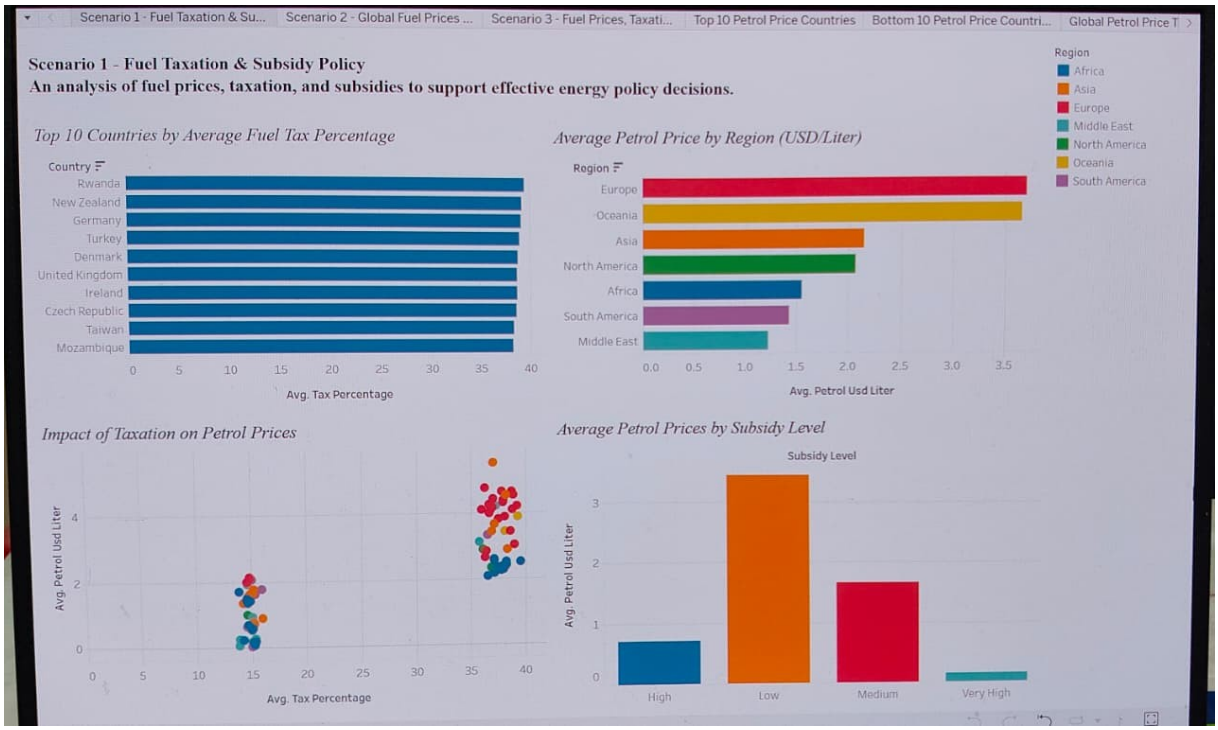
Major outcomes

• Taxation, regional price, and subsidy comparisons appear in one view.

• The scatter plot supports exploration of association between taxation and petrol price, not causal conclusions.

• The regional legend identifies the colour assigned to each region.

Latest dashboard screenshot



Dashboard 2 — Scenario 2: Global Fuel Prices & Supply Chain Planning

Compare country-level fuel prices and review regional and historical trends relevant to supply-chain planning.

Layout and visualisation components

Chart / component	Visual type	Design purpose
Top 10 Countries by Average Fuel Price Compare	Stacked horizontal bar chart	Compare petrol and diesel measures for countries ranked in the top 10 by average petrol price.
Bottom 10 Countries by Average Fuel Price Compare	Stacked horizontal bar chart	Compare petrol and diesel measures for countries ranked in the bottom 10 by average petrol price.
Average Petrol Price by Region (USD/Liter)	Horizontal bar chart	Compare average petrol prices by region.
Brent Crude Oil vs. Regional Petrol Price Trends	Time-series line chart	Review historical Brent crude and regional petrol-price trends over the displayed

period.

Interactivity and layout

The dashboard displays a Region legend and a Measure Names legend for petrol and diesel. Country rankings are based on average petrol price, not a combined petrol-and-diesel ranking.

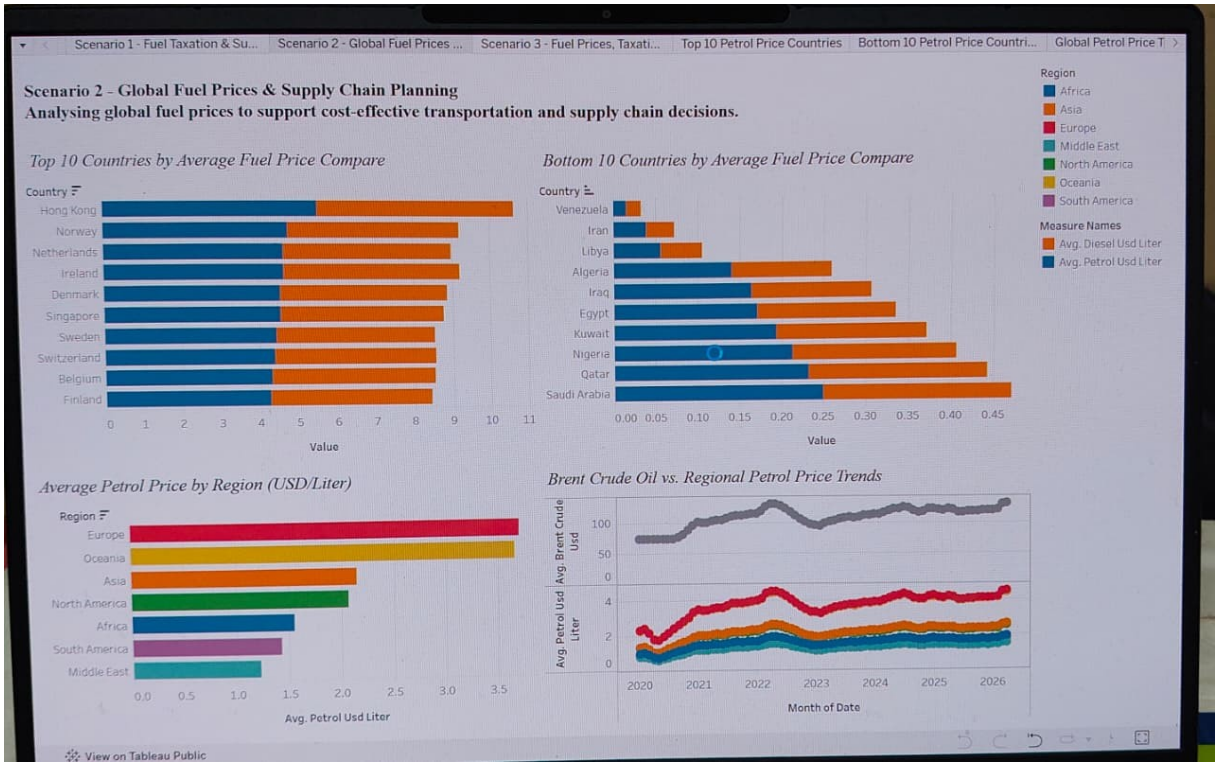
Major outcomes

• The ranking views support comparison of high- and low-price country groups.

• The regional chart summarizes geographic differences in average petrol prices.

• The time-series view places Brent crude and regional petrol-price trends together; it does not prove causation.

Latest dashboard screenshot



Dashboard 3 — Scenario 3: Fuel Prices, Taxation & Economic Development

Explore fuel prices, taxation, and subsidy patterns across economies with different income levels.

Layout and visualisation components

Chart / component	Visual type	Design purpose
Impact of Taxation on Petrol Prices	Scatter plot	Explore taxation and petrol-price patterns; the Income Level filter allows group-focused exploration.
Average Petrol Prices by Subsidy Level	Bar chart	Compare petrol prices by subsidy category with dashboard filters available.
Average Petrol Price by Income Level	Bar chart	Compare average petrol prices across low-, middle-, and high-income categories.
Average Fuel Tax Percentage by Income Level	Bar chart	Compare average fuel-tax percentages across income categories.

Interactivity and layout

Visible controls include Income Level, Region, and Subsidy Level filters, plus colour legends. Filters support exploration of subsets of the data.

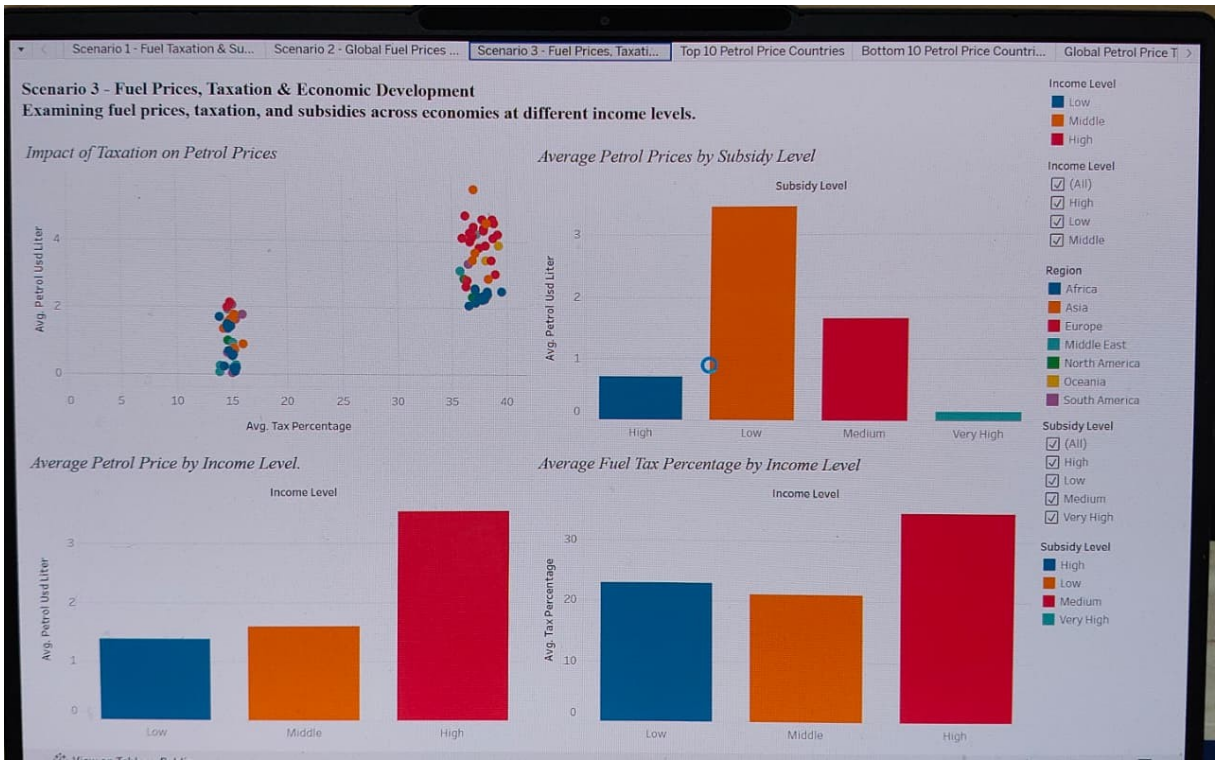
Major outcomes

• Average petrol prices and average tax percentages can be compared by income category.

• Filters support focused exploration by income level, region, and subsidy level.

• Group-level patterns are descriptive comparisons, not proof that income level causes a particular fuel price or tax rate.

Latest dashboard screenshot



Design decisions and limitations

• The three dashboards use a chart-grid layout, descriptive titles, and legends or filters where shown in the screenshots.

• Colour distinguishes regions, measures, income levels, and subsidy categories. Consult each displayed legend because colour meanings differ across dashboards.

• The screenshots show a desktop Tableau view. Mobile responsiveness has not been verified.

• Drill-down behaviour, custom icons, and a separate Tableau Story are not claimed as implemented in this document.

• The workbook uses the supplied historical CSV dataset and should not be described as a live fuel-price feed.

• No formal usability study or performance benchmark is claimed in this design document.

Activity 2 — Publish Dashboard on Tableau Public

Published Tableau Public workbook:

https://public.tableau.com/app/profile/harshvardhan.gupta1425/viz/Global_Fuel_Prices_and_Taxation_Trends_17932843176370/Scenario1-FuelTaxationSubsidyPolicy?publish=yes

The published workbook contains the three scenario dashboards. The link opens Scenario 1 first; users can navigate to the other dashboard tabs in the workbook.